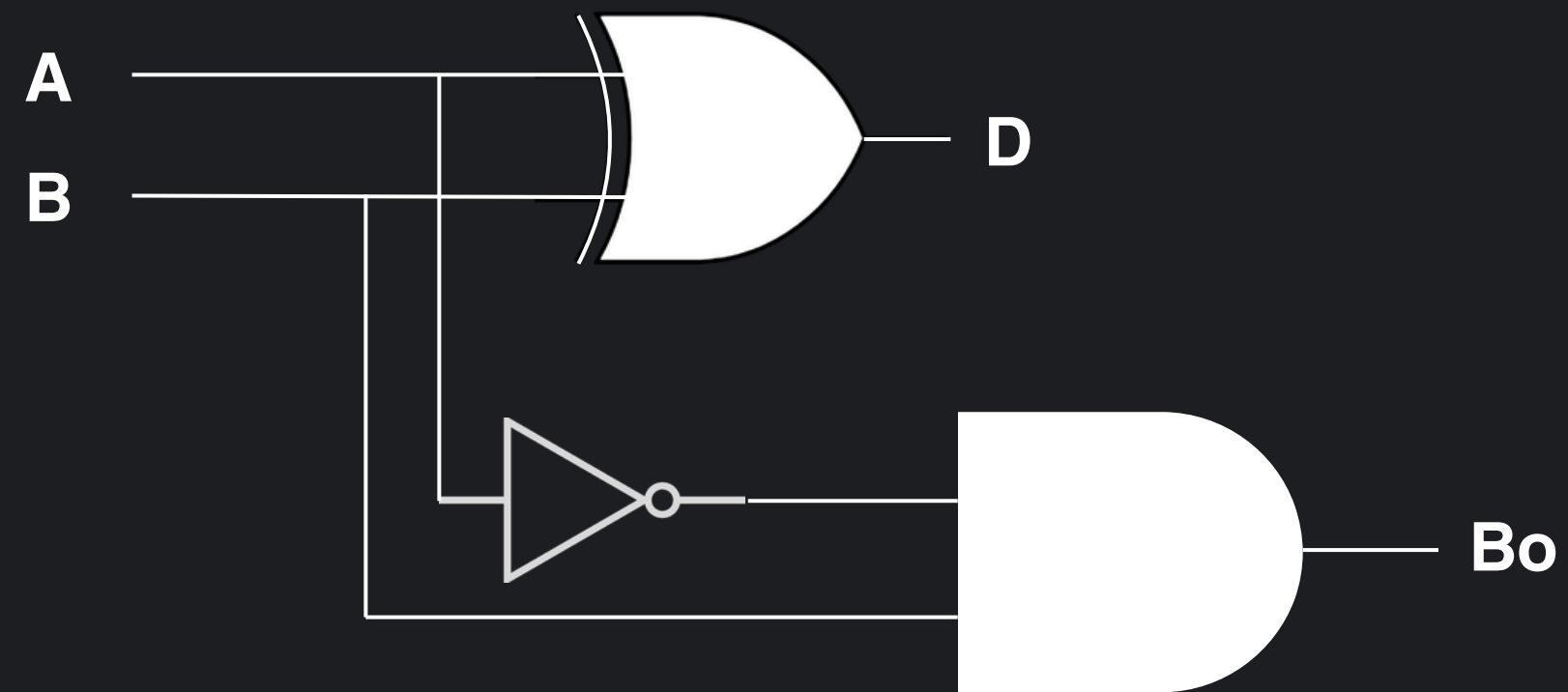


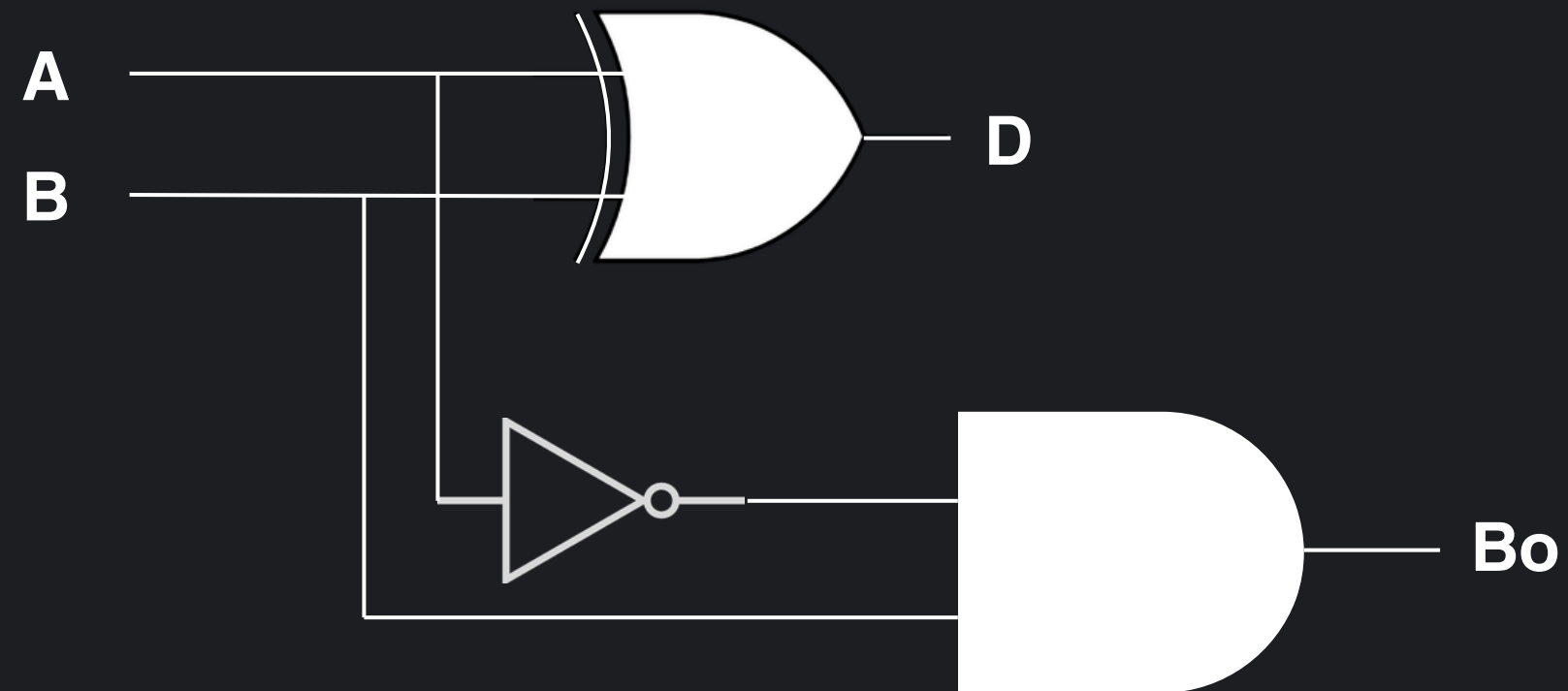
Half Subtractor



What is a Half Subtractor?

A Half Subtractor is a basic combinational logic element that performs the arithmetic subtraction of two binary bits. It produces two outputs, Difference and Borrow, based on its two inputs which consist of the minuend and the subtrahend.

Truth Table



A	B	D	Bo
0	0	0	0
0	1	1	1
1	0	1	0
1	1	0	0

Function of the Half Subtractor

A Half Subtractor performs a logical arithmetic subtraction on its two binary inputs. The main function of a Half Subtractor is to serve as a basic arithmetic element and produce an output signal (Difference and Borrow) according to this rule:

Output Difference = 1 when the inputs (A, B) are different (i.e., when one input is 1 and the other is 0).

Output Difference = 0 when the inputs are the same (i.e., both are 0 or both are 1).

Output Borrow = 1 only when the first input (A) is less than the second input (B) (specifically when $A=0$ and $B=1$).

Output Borrow = 0 in all other cases (when A is greater than or equal to B).